SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 3 – CODING CHALLENGE

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO1 – Imitation (L1)	**Assessment:**	Assessment Tool 3 – Coding Challenge
Weightage:	30% of CIA		
CO5 Statement:	Analyse NLP models, regular expression patterns, finite state automata, morphological processing techniques and to interpret language processing. (BL-3)		
Student Name:	G.Pranathi	**Reg. No.:**	192472237
Section / Batch:	CSE-AI/2024	**Date:**	

Code of Conduct

I G.PRANATHI [19247237] certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- Develop a complete and modular Python program that meets all the functional requirements specified in the coding challenge using appropriate algorithms, data structures, and programming practices.
- Test the program using multiple valid and invalid test cases, and clearly display the input, processing, and output to verify the correctness of the implementation.
- Follow standard coding practices by using meaningful variable names, proper indentation, comments, exception handling (where applicable), and upload the source code, sample outputs, and README file to the designated GitHub repository.
- Duration: 40 minutes.

Section / Topic	Focus	Q Nos
Regular Expression Pattern Matching	Regular Expressions, Basic Regular Expression Patterns	1
Finite State Automata Implementation	Finite State Automata, Models and Algorithms	2
Advanced Regular Expression Search	Regular Expressions, Basic Regular Expression Patterns	3

Annexure A – Coding Challenge

Section 1: Intelligent Email and Password Validator using Regular Expressions

1. **Problem Statement:**

A software company is developing a user registration system. Before creating an account, the system must validate user credentials based on predefined security policies.

Develop a Python application that validates:
Email Address
Password
Mobile Number

using Regular Expressions.

Validation Rules
Email
Must start with an alphabet.
May contain letters, digits, '.', '_' before '@'.
Domain should contain only alphabets.
Valid extensions: .com, .org, .edu, .net, .in
Password
Minimum 8 characters
At least one uppercase letter
At least one lowercase letter
At least one digit
At least one special character (@,#,$,%,&,!)
Mobile Number
Must contain exactly 10 digits.
First digit should be between 6–9.
Expected Output

Display:

Valid Email / Invalid Email
Strong Password / Weak Password
Valid Mobile Number / Invalid Mobile Number

Section 2: Design a Finite State Automata (FSA) Simulator

2. **Problem Statement**

Develop a Python-based Deterministic Finite Automata (DFA) simulator.

The program should:

- Accept the DFA description
    - States
    - Input alphabet
    - Transition table
    - Initial state
    - Final state(s)
- Accept multiple input strings.
- Simulate state transitions.
- Display whether the string is Accepted or Rejected.

Example

Language:
Strings ending with "ab"

Input
abaab

Output
Transition Path:
q0 → q1 → q0 → q1 → q2
Accepted

Section 3: Smart Pattern Matching Engine using Regular Expressions

3. Problem Statement

Develop a Python-based text search engine that performs pattern matching using Regular Expressions.

The system should support:
Word Search
Prefix Search
Suffix Search
Date Search
Phone Number Search
Hashtag Search
Mention (@username) Search

Example Input
Meeting on 12/09/2026
Call 9876543210
#NLP
@OpenAI
natural language processing

User Menu
1.Search Date
2.Search Phone Number
3.Search Hashtag
4.Search Mention
5.Search Prefix
6.Search Suffix

Expected Output
Display all matching patterns based on user selection.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Clearly understands problem constraints, edge cases, and competitive requirements	Understands problem with minor gaps in constraints	Partial understanding; misses key constraints	Misinterprets problem or constraints
Algorithm	25%	Selects optimal algorithm with best time and space complexity for competitive constraints	Uses efficient algorithm with minor scope for improvement	Uses basic/inefficient approach	No clear algorithmic approach
Code Implementation	20%	Writes correct, efficient, and clean code that passes all constraints	Code works with minor errors or inefficiencies	Partially correct code; fails for some cases	Code does not execute or gives incorrect results
Competitive Performance	20%	Solves problem within time limits and handles large inputs effectively	Solves within acceptable time with minor delays	Struggles with time limits or larger inputs	Unable to solve within time constraints
Test Case Handling	15%	Handles all test cases including edge and hidden cases	Handles most test cases	Misses important edge cases	Fails on multiple test cases

Q. No. / Criteria	Problem Understanding	Algorithm	Code Implementation	Competitive Performance	Test Case Handling	Sub. Total	Total %
1							
2							
3							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Section 1: Intelligent Email and Password Validator using Regular Expressions

Aim

To develop a Python program that validates an Email Address, Password, and Mobile Number using Regular Expressions.

Algorithm

1. Start the program.
2. Read the email, password, and mobile number from the user.
3. Validate the email using a regular expression.
4. Validate the password by checking:
    - Minimum 8 characters
    - At least one uppercase letter
    - At least one lowercase letter
    - At least one digit
    - At least one special character (@, #, $, %, &, !)
5. Validate the mobile number using a regular expression.
6. Display whether each input is valid or invalid.
7. Stop the program.

Code

```
import re

email = input("Email: ")

password = input("Password: ")
mobile = input("Mobile: ")

email_pattern = r'^[A-Za-z][A-Za-z0-9._]*@[A-Za-z]+\.(com|org|edu|net|in)$'
mobile_pattern = r'^[6-9][0-9]{9}$'

print("Valid Email" if re.fullmatch(email_pattern, email) else "Invalid Email")

if (len(password) >= 8 and
    re.search(r'[A-Z]', password) and
    re.search(r'[a-z]', password) and
```

```
    re.search(r'[0-9]', password) and
    re.search(r'[@#$%&!]', password)):
    print("Strong Password")
else:
    print("Weak Password")

print("Valid Mobile Number" if re.fullmatch(mobile_pattern, mobile)
```

Input

Email Address

Password

Mobile Number

Sample Input

Email : pranathi123@gmail.com

Password : Pranathi@123

Mobile : 9876543210

Output

Valid Email

Strong Password

Valid Mobile Number

Result

The Python program successfully validates the Email Address, Password, and Mobile Number using Regular Expressions and displays the appropriate validation results.

Section 2: Design a Finite State Automata (FSA) Simulator

Aim

To develop a Python program to simulate a **Deterministic Finite Automata (DFA)** that accepts strings ending with **"ab"**.

Algorithm

1. Start the program.
2. Define the DFA states, input alphabet, transition table, initial state, and final state.
3. Read the input string from the user.
4. Start from the initial state.

5. For each character in the input string:
    - Move to the next state using the transition table.
    - Store the transition path.
6. After processing the string, check whether the final state is an accepting state.
7. Display the transition path and print **Accepted** or **Rejected**.
8. Stop the program.

Code

```
# DFA for strings ending with "ab"

transitions = {

    'q0': {'a': 'q1', 'b': 'q0'},

    'q1': {'a': 'q1', 'b': 'q2'},

    'q2': {'a': 'q1', 'b': 'q0'}

}

start_state = 'q0'

final_state = 'q2'

string = input("Enter input string: ")

state = start_state

path = [state]

for ch in string:

    if ch not in ['a', 'b']:

        print("Invalid Input")

        exit()

    state = transitions[state][ch]
```

```
    path.append(state)
    print("Transition Path:")
    print(" → ".join(path))
    if state == final_state:
        print("Accepted")
    else:
  print("Rejected")
```

Input

Input String: abaab

Output

Transition Path:

q0 → q1 → q0 → q1 → q2

Accepted

Result

The DFA simulator successfully processes the input string, displays the transition path, and determines whether the string is **Accepted** or **Rejected**.

Section 3: Smart Pattern Matching Engine using Regular Expressions

Aim

To develop a Python-based text search engine that performs pattern matching using **Regular Expressions** for searching words, prefixes, suffixes, dates, phone numbers, hashtags, and mentions.

Algorithm

1. Start the program.
2. Read the text from the user.
3. Display the search menu.
4. Read the user's choice.
5. Use the corresponding Regular Expression to search the text.

Display all matching patterns.

6. Stop the program.

Code

```
import re

text = """Meeting on 12/09/2026

Call 9876543210

#NLP

@OpenAI

natural language processing"""

print("\n1.Word Search")

print("2.Prefix Search")

print("3.Suffix Search")

print("4.Date Search")

print("5.Phone Number Search")

print("6.Hashtag Search")

print("7.Mention Search")

choice = int(input("Enter your choice: "))

if choice == 1:

    word = input("Enter word: ")

    print(re.findall(r'\b' + word + r'\b', text))

elif choice == 2:
```

```
 pre = input("Enter prefix: ")
    print(re.findall(r'\b' + pre + r'\w*', text))
elif choice == 3:
    suf = input("Enter suffix: ")
    print(re.findall(r'\b\w*' + suf + r'\b', text))
elif choice == 4:
    print(re.findall(r'\b\d{2}/\d{2}/\d{4}\b', text))
elif choice == 5:
    print(re.findall(r'\b[6-9]\d{9}\b', text))
elif choice == 6:
    print(re.findall(r'#\w+', text))
elif choice == 7:
    print(re.findall(r'@\w+', text))
else:
print("Invalid Choice")
```

Input

Meeting on 12/09/2026
Call 9876543210
#NLP
@OpenAI
natural language processing

Output

Example (Choice = 2)

Phone Numbers Found:

9876543210

Result

The program successfully searches and displays the required patterns using Regular Expressions.